

Dysprosium-164

¹⁶⁴Dy (Dy I), nuclear spin $I = 0$. Energy levels and transitions below 2 μm : 39 levels, 55 lines.

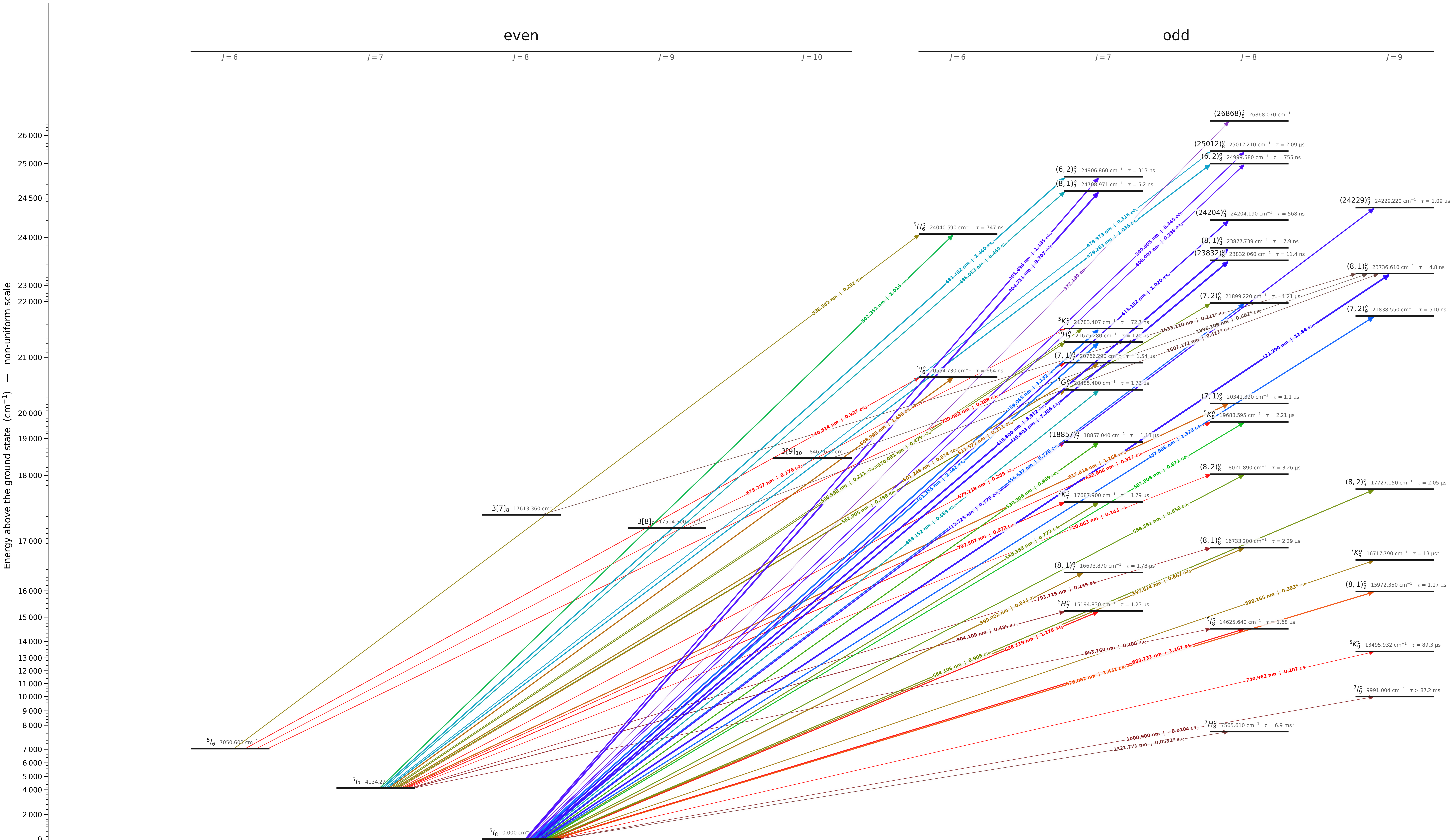
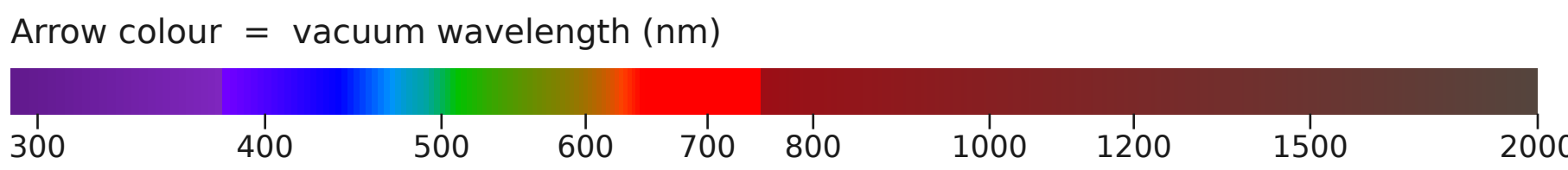
Dy⁺ ionisation limit 47901.76 cm⁻¹ = 5.93906 eV (208.761 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| e r \| J' \rangle|$ in ea

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U e r U \rangle $ (eao)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^5f_6 - (26868)^5g_6$	372.1890	372.0832	805.4845	-	-	-	-	UV ground-state line
$^5f_8 - (8, 1)^5g_6$	404.7113	404.5970	740.7563	9.707	1.920e+08	30.61	99.8	strong blue J=8->7 line
$^5f_6 - (8, 1)^5g_6$	418.8001	418.6821	715.8366	8.812	1.260e+08	20.15	99.5	strong blue J=8->8 line
$^5f_8 - (23832)^5g_6$	419.6028	419.4846	714.4672	7.386	8.800e+07	13.96	100	strong blue J=8->8 line
$^5f_6 - (8, 1)^5g_6$	421.2902	421.1715	711.6057	11.84	2.000e+08	33.16	100	main broad cooling line: Zeeman slower, transverse coolin...
$^5f_8 - (7, 2)^5g_6$	456.6373	456.5093	656.5221	0.726	6.600e+05	0.1315	79.9	pure 6s->6p reference line for King plots
$^5f_6 - (8, 1)^5g_6$	564.1065	563.9499	531.4466	0.908	4.900e+05	0.07764	100	closed J=8->9 narrow line
$^5f_8 - ^7K_6$	598.1652	597.9995	501.1867	0.393*	7.700e+04	0.01224	100	"598 nm" closed J=8->9 narrow line proposed for narrow-li...
$^5f_8 - (8, 1)^5g_6$	599.0223	598.8564	500.4696	0.944	5.600e+05	0.0891	99.7	narrow J=8->7 probe line
$^5f_8 - (8, 1)^5g_6$	626.0819	625.9088	478.8390	1.431	8.900e+05	0.136	100	narrow-line MOT
$^5f_8 - ^5H_6$	658.1186	657.9368	455.5295	1.275	7.700e+05	0.1294	94.7	narrow probe line used for Doppler thermometry of the 421...
$^5f_8 - ^5H_6$	683.7308	683.5421	438.4657	1.257	5.930e+05	0.09474	99	narrow J=8->8 optical pumping / demagnetization cooling l...
$^5f_8 - ^5K_6$	740.9619	740.7579	404.5990(10)	0.207	1.120e+04	0.001782	100	ultra-narrow-line MOT / narrow-line cooling in tweezers
$^5f_8 - ^7I_6$	1000.9004	1000.6261	299.5228	0.010	1.150e+01	1.825e-06	100	ultra-narrow
$^5f_8 - ^7H_6$	1321.7705	1321.4091	226.8113	0.053*	1.460e+02	2.307e-05	100	proposed ultra-narrow telecom-band line

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^5f_6 - (26868)^o_8$	372.189	164-163	-44(10)
$^5f_8 - (26868)^o_8$	372.189	164-162	-178(13)
$^5f_6 - (26868)^o_8$	372.189	164-161	-194(13)
$^5f_8 - (26868)^o_8$	372.189	164-160	-339(13)
$^5f_8 - (8.1)^o_9$	421.290	164-163	-616.3(5)
$^5f_6 - (8.1)^o_9$	421.290	164-162	-913.2(8)
$^5f_8 - (8.1)^o_9$	421.290	164-161	-1635.0(10)
$^5f_6 - (8.1)^o_9$	421.290	164-160	-1895.0(20)
$^5f_6 - (8.1)^o_9$	421.290	164-158	-2868(9)
$^5f_8 - (8.1)^o_9$	421.290	164-156	-4300(15)
$^5f_6 - (7.2)^o_8$	456.637	164-163	-660(3)
$^5f_8 - (7.2)^o_8$	456.637	164-162	-971.0(20)
$^5f_6 - (7.2)^o_8$	456.637	164-161	-1744(3)
$^5f_8 - (7.2)^o_8$	456.637	164-160	-2020(3)
$^5f_6 - (7.2)^o_8$	456.637	164-158	-3061(4)
$^5f_8 - (7.2)^o_8$	456.637	164-156	-4604(5)
$^5f_8 - (8.2)^o_9$	564.106	164-163	-685.0(20)
$^5f_6 - (8.2)^o_9$	564.106	163-162	-328(3)
$^5f_6 - (8.2)^o_9$	564.106	162-161	-801(3)
$^5f_8 - (8.2)^o_9$	564.106	161-160	-284.0(20)
$^5f_6 - (8.2)^o_9$	564.106	164-162	-1012.0(20)
$^5f_8 - (8.2)^o_9$	564.106	162-160	-1086(5)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$